Solar Filament Instance Segmentation under Multi-Annotator Panoptic Evaluation

Anonymous Participant
IEEE BigData Cup 2026 – Solar Filament Segmentation

TABLE I
UNION ORACLE BY PASSES PER IMAGE.

passes/image	images	oracle PQ
1	66	1.000
2	16	0.733
3	24	0.601
all	106	0.782

TABLE II
LOCAL VALIDATION PQ (MUTUALLY EXCLUSIVE MASKS).

stage	PQ	SQ	RQ
YOLO only	0.366	0.649	0.564
+ U-Net refine	**0.399**	0.670	0.595

***Abstract*—We study instance segmentation of solar filaments in 2048×2048 GONG H-alpha observations under the organiser's panoptic quality metric. We show that the 707 training images contain 1,154 annotator passes scored independently, that cross-annotator agreement bounds PQ at 0.320, and that the union-of-annotators oracle reaches 0.782 but only 0.601 on three-pass images. Our pipeline couples a YOLO11-seg detector trained on union targets with a U-Net semantic prior that re-thresholds each proposal; it improves local PQ from 0.366 to 0.399 and transfers to the public leaderboard (0.290→0.340).**

I. TASK AND METRIC

Given a 2048×2048 GONG H-alpha image, predict one mask per filament. The metric is panoptic quality with IoU>0.5 matching,

$$\mathrm{PQ} = \frac{\sum_{(g,p)\in TP} \mathrm{IoU}(g,p)}{|TP| + 0.5|FP| + 0.5|FN|}, \tag{1}$$

evaluated per annotator pass and pooled. Submissions are CSV files of (`filament_id`, `segmentation_rle`); masks of one image may not overlap.

II. WHAT THE DATA CONTAINS

The training images carry **1,154 annotator passes**: 411 images with one pass, 145 with two, 151 with three; 8,199 annotations. Because each pass is scored independently, a filament marked by one expert is a TP for that pass and an FP for every other pass of the same image, and FP and FN carry equal weight. Scoring one expert's labels against another's gives **PQ 0.320** (SQ 0.631, RQ 0.507). Using the union of all expert instances as a perfect prediction bounds any method. On our held-out split this oracle scores **0.782**, but the bound depends strongly on annotation count: Any split-to-split comparison is therefore only meaningful at fixed composition, and a public/private gap is expected when the public half has more multi-pass images.

III. METHOD

Union targets. Duplicate instances across passes of one physical image are merged (mask IoU>0.6, 6,941→5,810 instances), so the detector learns to cover every filament any expert considered real. Training on single-pass labels raises YOLO's mask mAP50 from 0.52 to 0.68 yet *lowers* PQ (0.370 vs 0.366): it buys precision with recall.

Detector. YOLO11s-seg at 1280 px, 100 epochs, mosaic and scale augmentation, no colour jitter.

Semantic prior and refinement. A U-Net (resnet34, 1024 px, union targets) predicts a full-resolution probability field. Each proposal is re-thresholded against that field inside a dilated 6 px ROI; components below 200 px fall back to the original proposal. This moves boundaries toward image evidence without inventing or deleting instances: **Mutual exclusion.** Instances are assigned greedily by confidence; this constraint alone costs ≈0.018 PQ and is applied in every number reported here.

IV. NEGATIVE RESULTS

Ten further ideas did not help: single-annotator labels (0.370), merging U-Net components (0.348), flip TTA (0.371), 1536 inference from a 1280 model (0.391), YOLO11m (0.396), s+m ensemble (0.373), tiled 1024 inference (0.320), proposal re-ranking (no held-out gain), 200-epoch training (0.393), and contrast channels (0.387). In every case detection metrics rose while PQ did not.

V. ERROR ANALYSIS AND LIMITS

Splitting validation by annotation count locates the bottleneck: Mask quality is flat (SQ≈0.67); single-pass images lose 53% of instances to missed detections. Missed instances have mean best-IoU 0.273: they are found but their boundary falls just short of the 0.5 threshold. Recall is flat across radial position (72–76%), so limb darkening is not the cause.

TABLE III
MEASURED PQ BY ANNOTATOR COUNT.

passes	oracle	measured	SQ	RQ
1 (66)	1.000	0.377	0.672	0.561
2 (16)	0.733	0.370	0.665	0.556
3 (24)	0.601	0.437	0.670	0.652

VI. CONCLUSION

Local validation PQ is **0.399**; public leaderboard **0.340**. Under the multi-annotator protocol the achievable ceiling on this split is 0.782, and the binding constraints are 707 training images, thin low-contrast targets, and FP/FN symmetry that punishes both over- and under-prediction. U-Net-guided proposal refinement is the one component that transfers to the leaderboard.